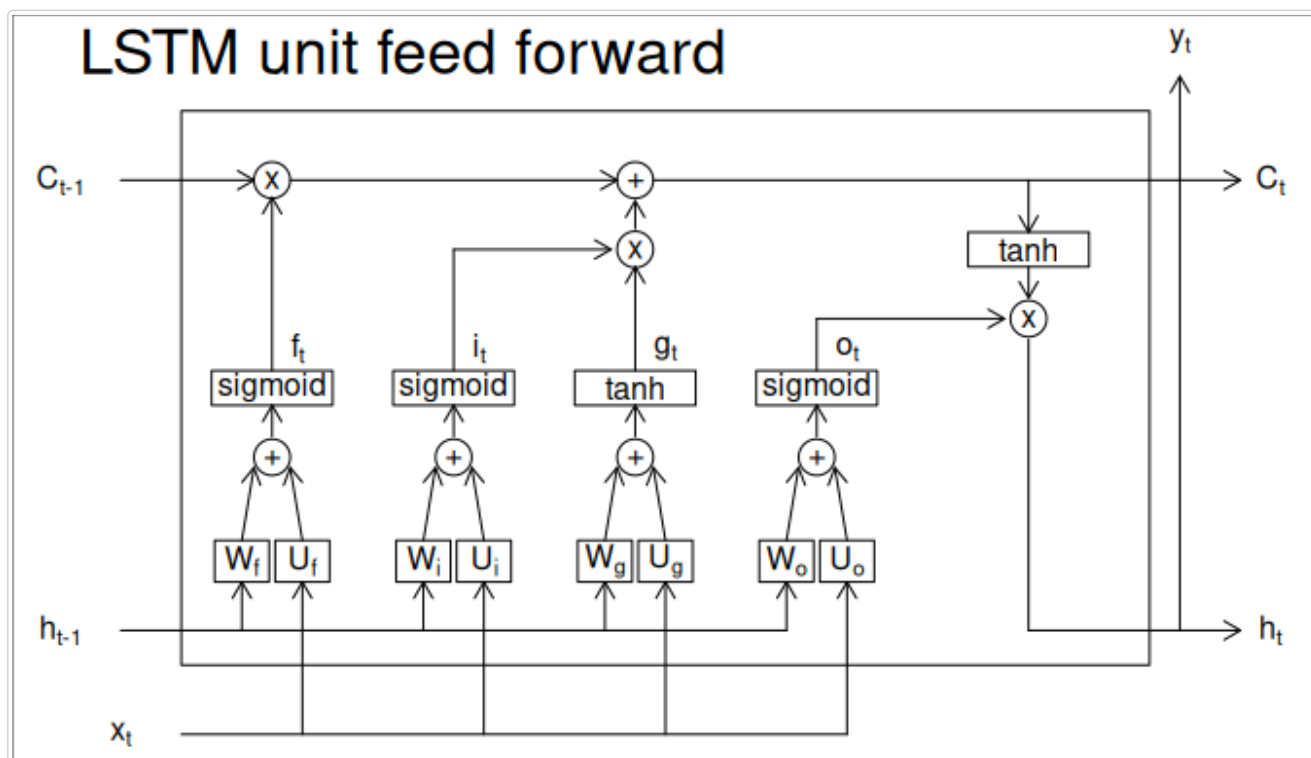


LSTM units

Dimitri Fichou

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Feed forward pass



$$f_t = \text{sigmoid}(h_{t-1} * W_f + x_t * U_f)$$

$$i_t = \text{sigmoid}(h_{t-1} * W_i + x_t * U_i)$$

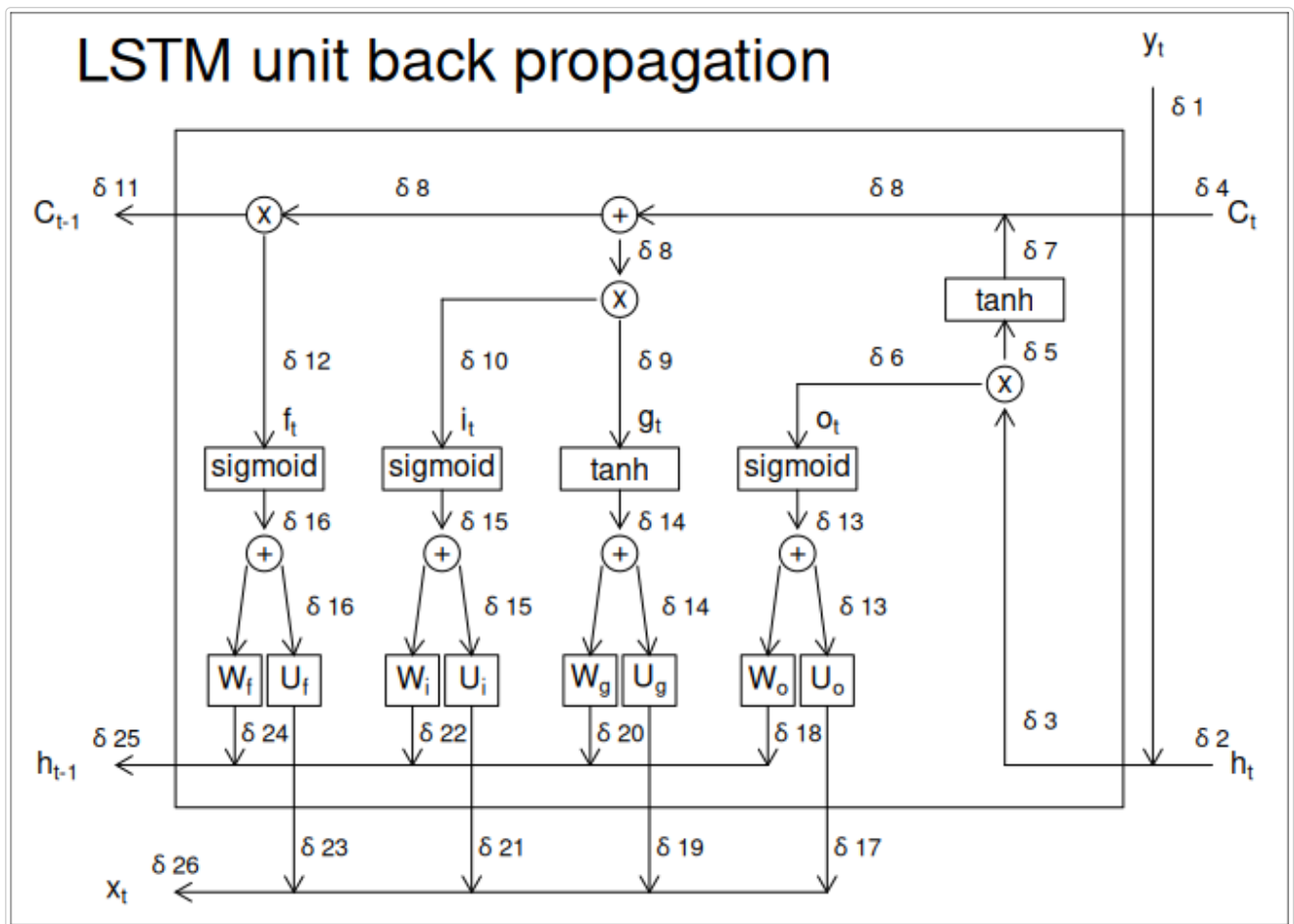
$$g_t = \text{tanh}(h_{t-1} * W_g + x_t * U_g)$$

$$o_t = \text{sigmoid}(h_{t-1} * W_o + x_t * U_o)$$

$$C_t = C_{t-1} \cdot f_t + i_t \cdot g_t$$

$$h_t = y_t = \text{tanh}(C_t) \cdot o_t$$

Back propagation pass



To perform the BPTT with a LSTM unit, we have the error coming from the top layer ($\delta 1$), the future cell ($\delta 4$), the future hidden state ($\delta 2$). Also, we have stored during the feed forward the states at each step of the feeding. In the case of the future layer, this error is just set to zero if not calculated yet. For convention, $\cdot$ correspond to point wise multiplication, while $*$ correspond to matrix multiplication.

The rules on how to back propagate come from [this post](#).

$$\delta 3 = \delta 1 + \delta 2$$

$$\delta 5 = \delta 3 \cdot 6 = \delta 3 \cdot o_t$$

$$\delta 6 = \delta 3 \cdot 5 = \delta 3 \cdot \tanh(c_t)$$

$$\delta 7 = \delta 5 \cdot f'(5) = \delta 5 \cdot \tanh'(\tanh(c_t))$$

$$\delta 8 = \delta 7 \cdot \delta 4$$

$$\delta 9 = \delta 8 \cdot 10 = \delta 8 \cdot i_t$$

$$\delta 10 = \delta 8 \cdot 9 = \delta 8 \cdot g_t$$

$$\delta 11 = \delta 8 \cdot 12 = \delta 8 \cdot f_t$$

$$\delta 12 = \delta 8 \cdot 11 = \delta 8 \cdot C_{t-1}$$

$$\delta 13 = \delta 6 \cdot f'(6) = \delta 6 \cdot \text{sigmoid}'(o_t)$$

$$\delta 14 = \delta 9 \cdot f'(9) = \delta 9 \cdot \tanh'(g_t)$$

$$\delta 15 = \delta 10 \cdot f'(10) = \delta 10 \cdot \text{sigmoid}'(i_t)$$

$$\delta 16 = \delta 12 \cdot f'(12) = \delta 12 \cdot \text{sigmoid}'(f_t)$$

$$\delta_{17} = \delta_{13} * U_o^T$$

$$\delta_{19} = \delta_{14} * U_g^T$$

$$\delta_{21} = \delta_{15} * U_i^T$$

$$\delta_{23} = \delta_{16} * W_f^T$$

$$\delta_{18} = \delta_{13} * W_o^T$$

$$\delta_{20} = \delta_{14} * W_g^T$$

$$\delta_{22} = \delta_{16} * W_i^T$$

$$\delta_{24} = \delta_{16} * W_f^T$$

$$\delta_{25} = \delta_{18} + \delta_{20} + \delta_{22} + \delta_{24}$$

$$\delta_{26} = \delta_{17} + \delta_{19} + \delta_{21} + \delta_{23}$$

The error δ_{11} , δ_{25} and δ_{26} are used for the next layers. Once all those errors are available, it is possible to calculate the weight update.

$$\delta W_f = \delta W_f + h_{t-1}^T * \delta_{16}$$

$$\delta U_f = \delta U_f + x_t^T * \delta_{16}$$

$$\delta W_i = \delta W_i + h_{t-1}^T * \delta_{15}$$

$$\delta U_i = \delta U_i + x_t^T * \delta_{15}$$

$$\delta W_g = \delta W_g + h_{t-1}^T * \delta_{14}$$

$$\delta U_g = \delta U_g + x_t^T * \delta_{14}$$

$$\delta W_o = \delta W_o + h_{t-1}^T * \delta_{13}$$

$$\delta U_o = \delta U_o + x_t^T * \delta_{13}$$